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RESEARCH ARTICLE

A Multimodal Deep Learning Framework Using ResNet-101 and Firefly-Based Feature Selection for Early Diagnosis of Dementia and Alzheimer's Disease

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ABSTRACT Dementia and Alzheimer's Diseases (AD) has global health challenges especially due to the progressive nature and the impact these diseases put on elderly population. Early detection is vital to improve prognosis and initiate timely intervention so that lives can be saved. This paper proposes a novel multimodal diagnostic framework that integrates clinical data and MRI imaging for the early classification of dementia and AD. The clinical pathway employs the OASIS dataset; wherein numerical features undergo statistical filtering followed by metaheuristic feature selection using the Firefly Algorithm. For imaging-based diagnosis, a deep Convolutional Neural Network (CNN) architecture is developed using MRI data, further optimized via a hybrid ResNet101–Support Vector Machine (SVM) model to enhance classification accuracy. The decision-level fusion module integrates outputs from both modalities to mitigate inter-class confusion and enhance diagnostic precision. When benchmarked against state-of-the-art models, the proposed framework achieves superior performance with an Accuracy of 0.933 and F1-Score of 0.921, highlighting its effectiveness and reliability. The results underscore the significance of integrated optimization and hybrid decision logic in advancing early-stage Alzheimer's diagnosis.

INDEX TERMS Alzheimer's diseases, firefly algorithm, deep neural network, machine learning.

I. INTRODUCTION

Dementia and Alzheimer's Diseases (AD) are global health challenges due to their progressive impact on the daily living activity among elder population. Dementia is defined as a clinical syndrome that involves memory loss, impairment in abilities cognitively and disturbances in behavior that can distort the daily routine of an elder human being. AD which is responsible for approximately 60-70% of dementia cases, is marked by neurodegenerative changes including intracellular tau protein tangles [1]. According to World Health Organization (WHO), about 47 million individuals globally suffers from dementia. WHO shows concerns and suggests that if it is not controlled, it might increase to about 82 million by 2030 due to the demographic trends of aging population worldwide [2]. That's why early detection of dementia and AD is crucial for the improvement of the patient outcomes as timely detection can significantly delay the cognitive impairments and may cure the patient as well. Large-scale initiatives such as the Alzheimer's Disease Neuroimaging Initiative (ADNI) have created standardised multi-centre neuroimaging datasets and protocols that provide a global reference for early detection research in Alzheimer's isolated from other forms of dementia [3]. Traditions diagnostic methods such as Positron Emission Tomography (PET), cerebrospinal fluid (CSF) biomarkers face significant limitations like high diagnostic costs, milted accessibility and the variation in disease presentation across different individuals. Manual interpretation of neuroimaging and clinical data for diagnosing dementia and AD has considerable challenges as well.

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The process is dependent upon expert knowledge majorly and it is prone to human errors [4].

In response to the challenges, the application of Artificial Intelligence (AI), primarily deep learning (DL) methods have emerged as a transformative solution in this diagnostic [5]. DL models like Convolution Neural Network (CNN) and autoencoders have demonstrated high performance in automated extraction and analysis of complex pattern from high dimensional medical datasets including MRI, PET and clinical data records [6]. Although pure deep learning models have gained significant success over traditional ML approaches, hybrid approach integrating DL with ML algorithms have recently attracted considerable interest due to their more accurate predictive performance and computational efficiency. Combination of ResNet101 models with Support Vector Machine (SVM) classifiers or CNN architectures modified by traditional feature selection and dimensionality reduction methods, have shown improved diagnostic accuracy over unimodal approaches [4], [7]. Notably, recent studies leveraging hybrid techniques demonstrated substantial improvements in diagnosing dementia and Alzheimer's compared to conventional methods, thus representing a promising direction for further research. Inspired by this advancement, this research introduces an innovative multimodal approach designed for early and accurate diagnosis of dementia and AD [8], [9].

The proposed framework has two districts yet interconnected sections. The first section focus on dementia detail using a technical indicator dataset extracted from clinical and numerical parameters (OASIS dataset) that employs Firefly Algorithm for optimized feature selection [10]. The second section focuses on AD classification using MRI imaging dataset. For this section, an advanced hybrid methodology has been developed combining ResNet101 with traditional ML classifier [11]. The integration of these two sections produces a sequential pathway of diagnosis upon positive detection of dementia followed by specific AD category is illustrated in Figure 1.

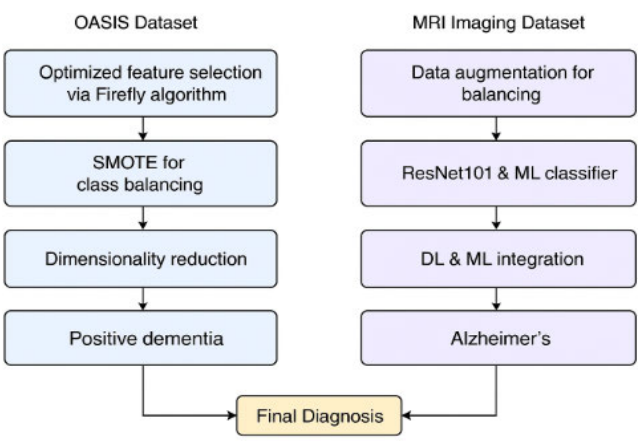

FIGURE 1. Integration of diagnostic workflow.

The proposed multimodal framework incorporates robust pre-processing that includes Synthetic Minority Over Sampling Technique (SMOTE) for dataset balancing and data augmentation thereby addressing inherent data challenges [12]. The main contributions of the proposed work are as follows:

For OASIS Dataset

a) Optimized feature selection using Firefly algorithm applied to clinical numeral indicators to enhance dementia detection.
b) Application of SMOTE to balance the dataset addressing class imbalance issue
c) Application of dimensionality reduction to represent high dimensional clinical data efficiently.

For MRI imaging dataset

a) balancing through sophisticated data augmentation technique increasing minority class instances for improved training
b) Development of a hybrid classification approach combining ResNet101 with traditional ML classifier.
c) Generalisation capability enfacement by integrating DL and ML models providing support to clinicians and medical expert.

The final diagnosis is concluded by integrating both modules, ensuring that dementia positive cases are further investigated for specific ALZHEIMER'S classification [13], [14]. The rest of the paper is organized in the following manner. Section II presents the literature review of the recent state of art techniques that represents common approaches. Section III is for the proposed work which includes the description of the dataset along with the proposed algorithm for feature selection from OASIS dataset, MRI dataset processing and the fusion approach.

II. LITERATURE SURVEY

Numerous research studies have explored Alzheimer's disease (AD) diagnosis through unimodal, hybrid, and multimodal learning paradigms. Islam [15] outlined the attributes of the OASIS dataset which serve as a foundation for feature analysis and preprocessing. Archana and Kalirajan [16] applied convolutional neural networks (CNNs) to classify AD using MRI data. Farooq et al. [17] developed a deep CNN-based multi-class classification approach using MRI images. Liu et al. [18] proposed a cascaded CNN structure combining multiple imaging modalities to enhance diagnostic performance. Khan et al. [19] incorporated transfer learning with intelligent data selection to improve MRI-based AD classification, while Mehmood et al. [20] introduced a Siamese CNN to discriminate between multiple AD stages. Gasmi [21] proposed an optimized hybrid framework using adaptive weight selection, advancing early AD detection. Ghazal and Issa [22] leveraged transfer learning for AD classification, showcasing the benefit of deep learning in medical imaging. Dar [23] suggested a CNN-based model to classify various AD stages using MRI images. Shafi et al. [24] worked on lung cancer detection, though not directly AD-related,

it demonstrates domain-relevant hybrid deep architectures. Islam and Zhang [25] proposed an ensemble of CNNs for brain MRI analysis, and Abrol et al. [26] employed residual learning for AD progression prediction. Khatun et al. [27] combined CNN and LSTM to enhance AD diagnosis from MRI images. Hazarika et al. [28] adopted a DNN for classifying AD stages based on MRI, and Fu'adah et al. [29] used the AlexNet CNN model to perform automatic AD classification. Razzak et al. [30] designed an ensemble-based system named PartialNet to detect AD using multi-resolutional MRI inputs. Pan et al. [31] introduced a CNN with ensemble learning to enable early detection. Rallabandi and Seetharaman [32] applied MRI-PET fusion to classify AD stages using deep learning, a strategy that aligns with the dual-path fusion logic adopted in the proposed work. Previous work has demonstrated that deep ensemble learning methods employing multimodal data can significantly improve diagnostic accuracy when seeking to detect brain diseases [33]. Zuo et al. [34] presented adversarial hypergraph fusion for early AD prediction, while Chatzianastasis et al. [35] explored neural architecture search with multimodal fusion for dementia diagnosis. Liu et al. [18] previously explored cascaded CNN models, reinforcing the importance of combined modalities. These works provide a comprehensive foundation and justify the proposed integration of feature-optimized clinical learning and hybrid deep image analysis for accurate and early-stage AD diagnosis. Islam [15] provided foundational insights into the attributes of the OASIS dataset, serving as a reference for preprocessing and feature analysis. Archana and Kalirajan [16] utilized CNN models over MRI datasets for Alzheimer classification, achieving moderate success in binary classification settings. Similarly, Farooq et al. [17] demonstrated the application of deep CNNs for multi-class AD detection, emphasizing the role of transfer learning. Liu et al. [18] extended this by proposing cascaded multimodal CNN architectures, integrating MRI and PET data for enriched prediction. Rallabandi and Seetharaman [32] employed deep learning-based fusion of MRI-PET imaging, closely aligning with the fusion logic of the proposed work, though lacking optimized feature selection mechanisms. Similarly, a multi-task framework for deep learning has been used for clinical diagnostics by combining imaging and non-imaging biomarkers of disease for improved classification of Alzheimer's cases [36]. Khatun et al. [27] introduced a hybrid CNN-LSTM architecture, reflecting the growing interest in temporal modeling, whereas Mehmood et al. [20] proposed a Siamese CNN for classifying different AD stages with notable improvement in precision. Gasmi [21] integrated adaptive weight selection with hybrid deep learning, further validating the role of optimization layers. Dar et al. [23] built a CNN for stage-wise AD classification and achieved respectable accuracy (0.891), while Fu'adah et al. [29] used AlexNet for automatic MRI classification. Chatzianastasis et al. [35] and Zuo et al. [34] explored neural architecture search and adversarial hypergraph fusion respectively, introducing advanced multimodal learning strategies that align with the modular fusion concept of the current study.

A. GAP ANALYSIS AND POSITIONING OF THE PROPOSED WORK

Even though the previously listed works provide evidence of progress in AD diagnosis, they collectively identify gaps in the research. Many of the unimodal approaches [16], [17], [18], [19], [20], [23], [29] used only either imaging or clinical data which have restricted capability to interpret the full spectrum of the disease. The hybrid and multi-modal studies [18], [27], [32], [35] show improved performance as these approaches integrated modalities, however, most use feature-level fusion without specific methods to accommodate missing or unavailable modalities lowering real-world robustness.

On the other hand, our proposed framework combines Firefly Algorithm-optimized clinical features with ResNet101-based MRI assessment using a decision-level fusion to combine clinical input and MRI output explicitly designed to deal with one modality not being available. In addition, we demonstrate improved diagnostic capability when employing a two-tier SVM to classify the MRI outputs for detecting disease stage, in particular closely related classes for Mild dementia and Moderate dementia have been previously poorly considered in other works. By positioning our proposed framework to fill in existing gaps, we also improve applied utility in a heterogeneous health system.

TABLE 1. Dataset and models.

Year	Dataset Used	Algorithms Applied	Outcome	References
(2017)	MRI	Deep CNN	Multi-class detection, effective on MRI	[17]
(2018)	MRI + PET	Cascaded Multimodal CNNs	Improved diagnosis via modality integration	[18]
(2018)	Brain MRI	Ensemble of CNNs	High accuracy using multiple CNN branches	[25]
(2020)	MRI	Siamese CNN	Better stage separation, precision improved	[20]
(2020)	Neuroimaging	Residual Learning	Progression prediction of AD	[26]
(2021)	Multi-modal (unspecified)	Adversarial Hypergraph Fusion	Effective feature representation and robustness	[33]
(2022)	CT Scan (Lung)	Deep Learning-Based Support Vector Network	Effective cross-domain hybrid model, relevant for SVM integration comparison	[24]
(2023)	MRI + PET	Fusion-based DL	Fused model outperforms single-path networks	[33]
(2023)	MRI + Other modalities	Neural Arch. Search + Fusion	Improved classification through optimized fusion	[35]
(2023)	MRI	CNN (AlexNet)	Moderate success in classification (binary AD)	[16]
(2024)	MRI	Hybrid DL with adaptive weight	Early-stage classification with improved accuracy	[21]

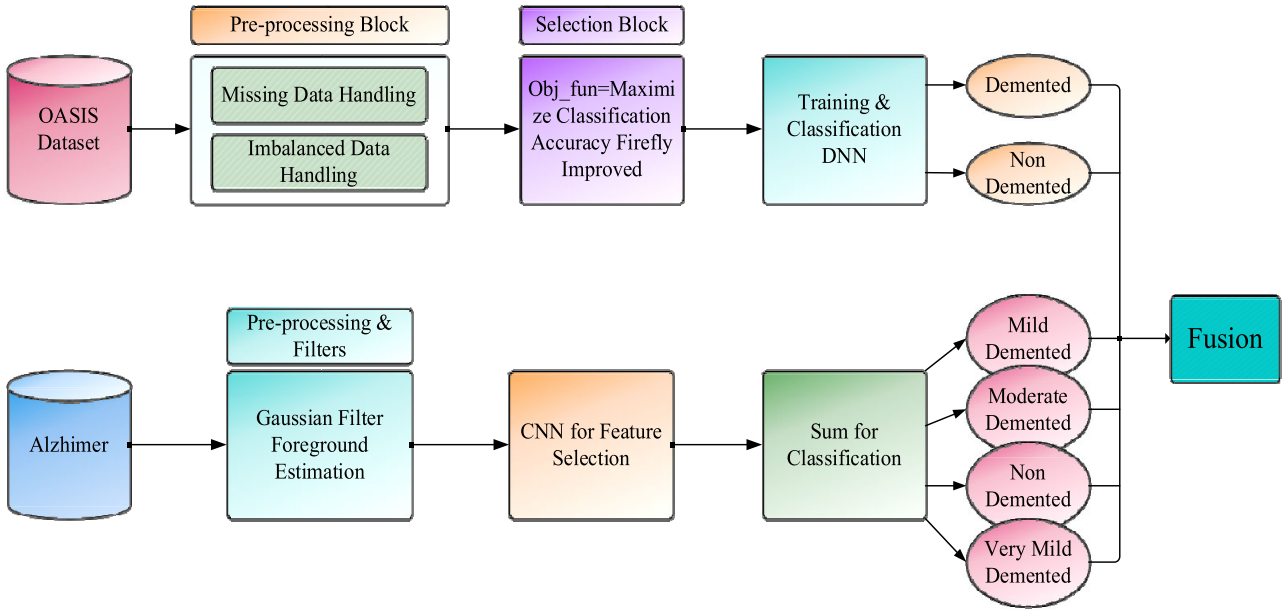

FIGURE 2. Work flow.

The proposed work is novel in three key respects: (i) the decision-level fusion framework has been specifically developed to handle the case where only one modality is available, (ii) the two-tier SVM structure allows the DLCFS method to break down challenging class boundaries for improved MRI-based staging and (iii) the hybrid ResNet101–SVM image classification integrated with optimized clinical features derived from the Firefly Algorithm. To our knowledge, this type of strategy to handle modality and other combined multimodal approaches has not been explored in any other dementia diagnosis studies.

III. MATERIALS AND METHODS

This section presents the multimodal diagnostic proposed framework for early detection of dementia and AD. The proposed framework has two parallel yet complementary pipelines namely for numerical clinical data (oasis dataset) and other for brain MRI IMAGE DATA (MRI Dataset) A fusion layer integrates both predictions. The overall work flow can be represented using Figure 2 as follows.

A. DATASET DESCRIPTION

The proposed work uses two distinct public available dataset and the description is as follows.

1) OASIS DATASET

The OASIS dataset is a widely used dataset for early detection of sign of dementia based on technical indicators or clinical parameters. This study utilized the OASIS dataset which is readily available at https://www.oasis-brains.org. It has 15 features from 373 subjects aged between 60 to 98 years. It also contains demographic details, clinical scores and brain volume indicators that are extracted from MRI scans.

TABLE 2. Features in OASIS dataset along with description.

No.	Feature Name	Description
1	Subject.ID	Unique identifier for each subject
2	MRI.ID	Unique identifier for each MRI scan
3	Group	Class label: Non-Demented, Demented, or Converted
4	Visit	Number of visits in the longitudinal study
5	MR.Delay	Time delay between scan and visit (can be unknown)
6	M.F	Gender: Male (M) or Female (F)
7	Hand	Dominant hand (usually all R)
8	Age	Age of the subject in years
9	EDUC	Years of education
10	SES	Socioeconomic status (1–5)
11	MMSE	Mini Mental State Examination score (0–30)
12	CDR	Clinical Dementia Rating (0 to 3)
13	eTIV	Estimated Total Intracranial Volume
14	nWBV	Normalized Whole Brain Volume
15	ASF	Atlas Scaling Factor

The primary label is calculated from clinical dementia Rating (CDR) scale. For label decoding, converted class records are reassigned based on their CDR values as follows.

$$y_i = \begin{cases} 0, & if\ CDR_i < 0.5 \\ 1, & if\ CDR_i \geq 0.5 \end{cases} \tag{1}$$

Eq(1) makes it a binary classification problem. Despite of having only two classes, the data in OASIS is highly

imbalanced which is resolved in the proposed work via synthetic oversampling. The features of the OASIS dataset is presented in Table 2.

2) MRI ALZHEIMER DATASET

The second dataset consists of 6400 pre-annotated MRI images, downloaded from Kaggle's Alzheimer dataset repository. The images are divided into four categories: Mild Dementia, Moderate Dementia, Very Mild Dementia, and Non-Demented.

These classes represent varying disease severity levels and form the backbone of the visual pipeline in our system. The imbalance among class frequencies is addressed through data augmentation in later stages. A sample of class-wise image distributions is as follows:

- Non-Demented: 3200 images
- Very Mild: 2240 images
- Mild: 896 images
- Moderate: 64 images

B. THE PREPROCESSING ARCHITECTURE

As mentioned earlier, both the datasets requires intensive preprocessing for the management of data imbalance in OASIS dataset and for the enhancement in the image quality for MRI dataset.

1) OASIS DATASET: CLEANING AND CLASS REBALANCING

Missing Values: The SES feature has 5% missing values and MMSE has 1% missing entries. The restoration has been done using the median of non-missing values.

$$x' = Median(x_{valid}) \tag{2}$$

Algorithm 1 Preprocessing of OASIS Dataset

Require: Clinical dataset D_{OASIS} with missing values and class imbalance
Ensure: Cleaned and balanced dataset D'_{OASIS}
1: Remove non-predictive columns: Subject.ID, MRI.ID, Hand
2: Impute missing values in **SES** and **MMSE** using column-wise median:

$$x' = Meian(x_{valid}) \quad \text{(Eq. 1)}$$

3: Transform class labels using CDR threshold:

$$y_i = \begin{cases} 0 & if\ CDR_i < 0.5 \\ 1 & if\ CDR_i \geq 0.5 \end{cases} \quad \text{(Eq. 2)}$$

4: Apply SMOTE to resolve class imbalance:

$$x_{new} = x + \lambda\,(x_{nn}zx)\,,\ \lambda \in [0, 1] \quad \text{(Eq. 3)}$$

5: Split data into training and testing sets (80% / 20%) **return** D'_{OASIS}

Feature such as "Hand", "Subject Id" and "MRI ID" were excluded due to low variance or non-predictive nature.

Class Imbalance Handling: The skewness in the dataset is quite high in the minority class. The proposed work handles it using SMOTE.

$$\boldsymbol{x_{new} = x + \lambda\,(x_{nn} - x)\,,\ \lambda \in [0, 1]} \tag{3}$$

Once the balancing is done, the dataset is divided into a ratio of 80-20 viz. 80% data is used for the training purpose whereas the rest of the data is used for the classification purpose.

2) MRI ALZHEIMER'S DATASET: FILTRATION AND CLEANING

Bias Field Correction: Non-uniformity in intensity due to patient positioning and scanner artifacts are corrected using

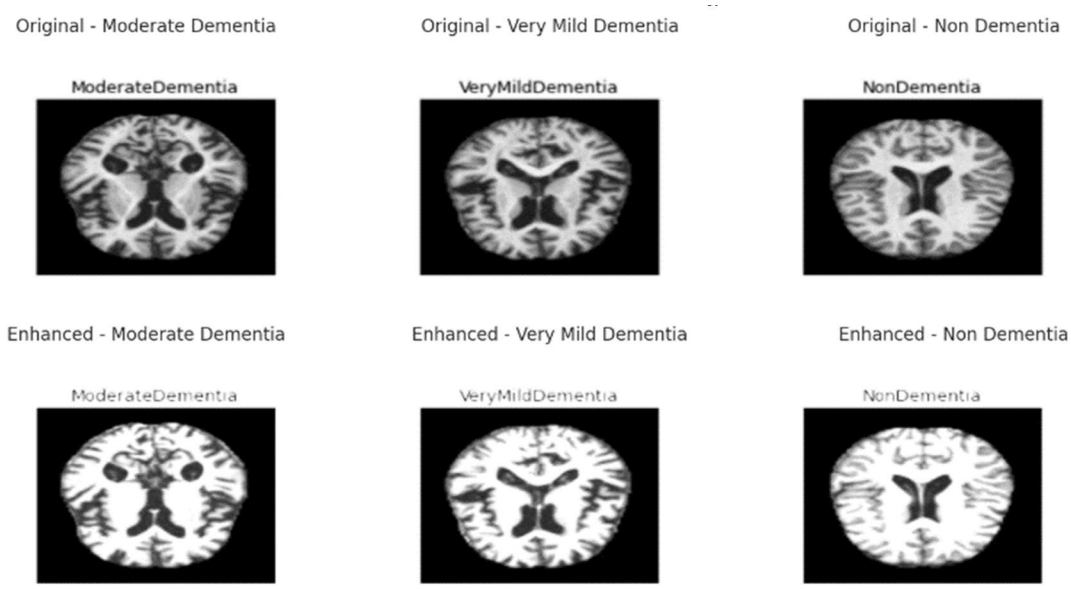

FIGURE 3. Outcome of processing.

histogram equalization and intensity normalization:

$$I' = \frac{I - \min(I)}{\max(I) - \min(I)} \tag{4}$$

Denoising and Smoothing: An averaging filter was applied to smoothen image textures and suppress pixel-level noise

$$\boldsymbol{I'} = \frac{\mathbf{1}}{\boldsymbol{N^2}} \sum\nolimits_{\boldsymbol{m,n \in N}} \boldsymbol{I_{i+m,j+n}} \tag{5}$$

Data Augmentation: To avoid overfitting and improve class representation, augmentation techniques such as rotation, flipping, brightness/contrast variation, and cropping were used during training:

$$I_{aug} = T_k(I), k = 1 \ldots .K \tag{6}$$

Algorithm 2 Preprocessing of MRI Dataset

Require: MRI image dataset D_{MRI} with bias field and class imbalance
Ensure: Enhanced and balanced dataset D'_{MRI}
1: **for all** image $I \in D_{MRI}$ **do**
2: Normalize intensity values:

$$I' = \frac{I - \min(I)}{\max(I) - \min(I)} \quad \text{(Eq. 4)}$$

3: Apply noise reduction via averaging filter:

$$I' = \frac{1}{N^2} \sum_{m,n \in N} I_{i+m,j+n} \quad \text{(Eq. 5)}$$

4: Store enhanced image I'
5: **end for**
6: Apply augmentation on minority classes using transformations:

$$I_{aug} = T_k(I), k = 1 \ldots K \quad \text{(Eq. 6)}$$

return D'_{MRI}

C. THE FEATURE SELECTION AND TRAINING ARCHITECTURE

The process of feature selection is applied to both the datasets namely the OASIS and the MRI dataset. For OASIS, firefly algorithm is used to select the most valuable technical indicators using the objective function to maximise the overall classification accuracy against the 4 classes. In the exact reverse order, the feature selection process for MRI dataset is made using CNN.

$$Obj_{fun_{oasis}} = argmax\,(CA)\,\forall i \in \{1 \ldots n\} \tag{7}$$

where n is total number of classes.

1) FEATURE SELECTION AND TRAINING MODULE FOR OASIS DATASET

To enhance the predictive performance and reduce overfitting, a two-stage feature selection strategy was applied to the OASIS dataset:

Step 1: Statistical Correlation Analysis

Initially, Pearson's correlation coefficient was calculated between each independent feature and the target class. Highly correlated features were prioritized, while those with redundancy or minimal correlation were removed to avoid collinearity.

Step 2: Firefly Algorithm-based Optimization

To further optimize the selected subset, the Firefly Algorithm (FA)—a meta-heuristic inspired by the flashing behaviour of fireflies—was employed. The fireflies' attractiveness and movement equations guide the search for an optimal subset of features that maximizes classifier accuracy.

Algorithm 3 Firefly Algorithm for Feature Selection

Require: Dataset D, number of fireflies N, maximum iterations T, distance coefficient γ, attractiveness β_o randomization factor α
Ensure: Optimal feature subset x∗ maximizing classification accuracy
1: Initialize population of fireflies x1, x2, …, $x_N \in \{0, 1\}$ d
2: **for** each firefly x_i **do**
3: Evaluate fitness using classifier accuracy:

$$F(x_i) = \text{Accuracy}(x_i) \quad \text{(Eq. 7)}$$

4: **end for**
5: for $t = 1$ to T **do**
6: for $i = 1$ to N do
7: for $j = 1$ to N do
8: if $f(x_j) > f(x_i)$**then**
9: Compute distance:

$$r_{ij} = \left|\left|x_i - x_j\right|\right|^2 \quad \text{(Eq. 8)}$$

10: Update position of firefly x_i:

$$x_i = x_i + \beta^0 e^{\left(-\gamma r_{ij}^2\right)(x_j - x_i)} + \alpha\varepsilon \quad \text{(Eq. 9)}$$

11: Apply sigmoid transformation to binarize features (if needed)
12: **end if**
13: **end for**
14: **end for**
15: Evaluate fitness of updated positions
16: **end for**
17: **return** firefly x* with the highest fitness

The Firefly Algorithm (FA) is a population-based meta-heuristic inspired by the bioluminescent communication behaviour of fireflies. It is particularly effective for solving combinatorial optimization problems such as feature subset selection. In the proposed methodology, FA is utilized to identify the most informative subset of features from the OASIS dataset that maximizes classification performance. Each firefly represents a binary vector, where each dimension corresponds to a feature. A value of '1' indicates the feature is selected. In our implementation, the Firefly Algorithm was set with a population size of 20 fireflies, a maximum of 50 iterations, light absorption coefficient $(\gamma) = 1.0$, attractiveness at zero distance $(\beta_o) = 1.0$, and randomization parameter $(\alpha) = 0.2$. The objective function is to maximize the classification accuracy of the DNN on validation data given the features selected. The pseudocode for the FA-based feature selection is given in Algorithm 3.

Once the features are selected, the features are passed to a three-layer deep neural network (DNN) with 32,16,8 neurons

at the respective 3 layers. The proposed DNN is trained using the Adam optimizer with a learning rate of 0.001, $\beta_o = 0.9$, and $\beta_o = 0.999$, unless stated otherwise. The hidden layers of the DNN employ the ReLU activation function, whereas the output layer uses Softmax. The networks are trained for 100 epochs, using mini-batches of 32, and monitored using early stopping to prevent overfitting.

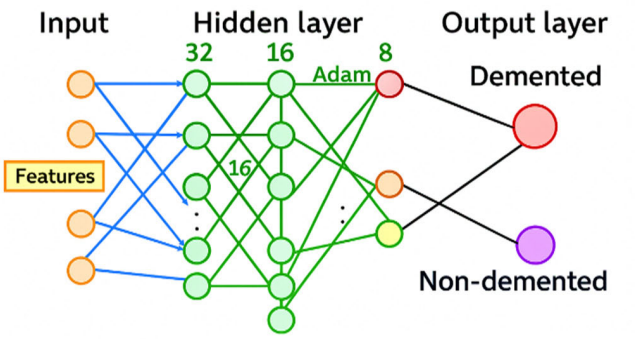

FIGURE 4. DNN Training for OASIS.

The outcome of DNN is the classified class $\mathbf{y_{num}}$ which is categorised in Demented and Non-Demented.

2) FEATURE SELECTION USING RESNET-101 AND TRAINING FOR MRI DATASET

The proposed uses ResNet-101 for the extraction of deep hierarchical features from pre-processed MRI images. ResNet-101 is a residual deep neural network with 101 layers. ResNet-101 deals with the vanishing gradient problem by introducing identity shortcut connection that gives the chance to the gradients to propagate more directly through the network.

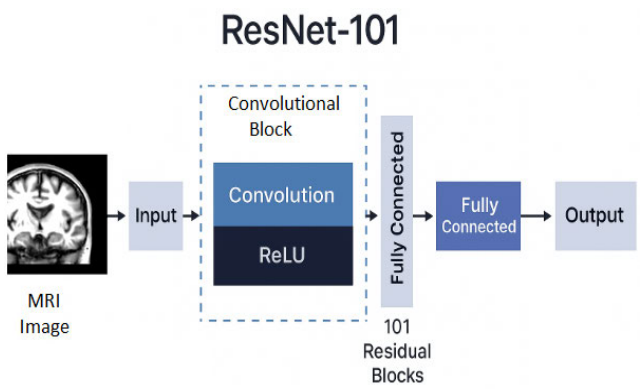

FIGURE 5. ResNet-101 fused model and individual modalities.

Let x denote the input MRI image and $F(x, W_i)$ denote the residual function learned by the $i - th$ block, where W_i represents the learnable weights in that block. A residual block in ResNet can be mathematically formulated as:

$$y_i = F\left(x_i, W_i\right) + x_i \tag{8}$$

where x is the input to the residual block, $F\left(x_i, W_i\right)$ is composed of two to three convolution layers and y_i is the output after residual addition.

The forward propagation through a sequence of residual blocks can be recursively defined as:

$$x_{i+1} = x_i + F(x_i, W_i) \tag{9}$$

The final output of the convolution block is denoted as f_{cnn}, it is flattened into a feature vector.

$$\hat{f} = Flatten\left(f_{cnn}\right) \in R^d \tag{10}$$

The flatten layer is passed to hybrid decision module and in case of proposed work, it is Support Vector Machine(SVM). The SVM classifier finds the hyperplane that maximizes the margin between the classes. For a binary classification, the decision function is given by

$$f\left(x\right) = sign\left(w^T \hat{z}(x) + b\right) \tag{11}$$

where w is the learned weight vector, b is the bias term and $f \wedge$ is the deep feature vector extracted from ResNet-101.

To improve interpretability and reduce classification complexity in the presence of four Alzheimer's disease stages (Non-Demented, Very Mild, Mild, and Moderate), the proposed work implements a two-tier classification strategy. Rather than solving a single 4-class problem, the system decomposes it into binary stages as follows:

Stage 1:

a) Classifier A distinguishes between Mild and Moderate dementia classes
b) Classifier B distinguishes between Non-Demented and Very Mild Demented

Stage 2:

The winner from the winners from Classifier A and Classifier B are passed to a final SVM classifier to predict the final diagnostic label.

It allows each SVM model to focus on sharper binary boundaries, improving overall performance. Mathematically, the final decision function is formulated as:

$$y_{img} = f_{stage2}(f_A\left(x\right), f_B(x)) \tag{12}$$

where $f_{A(x)}$ and $f_B(x)$ are the predicted outcome of classifier A and Classifier B which is SVM itself in case of proposed work. The classification process does not increase the computation complexity and hence it will not result into a significant increase in the overall process evaluation time but will result into a lighter classification mechanism for sure. Using an initial confusion matrix, the two-tier SVM architecture design could be rationalized as disambiguation in the form of misclassifications occurring mostly between Mild and Moderate dementia, and Non-Demented and Very Mild dementia based on the level of the current architecture study. Thus, these close-proximity classes were separated into two discrete binary classification tasks to help each SVM classify between classes with closer decision boundaries.

TABLE 3. Ablation study comparing Two-tier SVM vs. Single 4-class SVM on MRI Dataset.

Architecture	Accuracy	F1-Score
Single 4-class SVM	0.884	0.888
Two-tier SVM	0.900	0.908

This extension is consistent with clinical procedures in which difficulty in distinguishing the subsequent progression of individual stages of dementia is often more challenging from one to another, as compared to distinguishing an early stage from a substantive set of dementia behaviors. An ablation study (Table 3) confirmed that using a single 4-class SVM instead of a two-tier structure showed a reduction in accuracy from 0.900 to 0.884 and a reduction in F1-score from 0.908 to 0.888, ultimately confirming that the proposed decomposition improves classification performance.

D. FUSION

The final outcome for the proposed work is to fuse the outcome of both the modularity's. The merging of OASIS clinical data and MRI imaging, allows the model to take advantage of the synergistic strengths of each modality. Clinical features contain early non-imaging features like demographics, cognitive assessment scores (MMSE, CDR), volumetric measures, which may reflect dementia risk before overt structural changes in the brain. MRI features calculated through ResNet101, reflect spatial and morphological changes in brain regions associated with the progression of AD that clinical scores may not be able to reflect.

Making predictions based on the decision level fusion of modalities allows the framework to be resistant to modality-specific noise or missing data, avoids false negatives, and improves diagnostic confidence. It is quite common in hospital settings that either clinical or imaging data is missing at the time of evaluation. The fusion approach makes it more likely that the system will generate reliable predictions in cases where only partial data are available making it feasible for practice in a variety of health care contexts.

Hence the proposed work establishes decision level fusion of outputs from two independent branches. Each branch produces output $y_{num} \in \{0, 1\}$ and $y_{img} \in \{0, 1, 2, 3\}$. The algorithm operates by first checking the availability of both outputs. If both are present, it compares the predictions. If both OASIS and MRI modules indicate a Non-Demented class (0), the final decision is confidently labelled as Non-Demented. If the OASIS output is Demented (1) and the MRI output belongs to a valid Demented stage (1, 2, or 3), the system defers to the MRI model for specific dementia staging. In the case where OASIS suggests dementia (1) but MRI indicates Non-Demented (0), the final decision defaults to Non-Demented, assuming imaging is more definitive, but the case is flagged for clinical review due to conflict.

Algorithm 4 Intelligent Decision-Level Fusion of OASIS and MRI Output

Require: $y_{num} \in \{0, 1\}$ (from OASIS), $y_{img} \in \{0, 1, 2, 3\}$ (from MRI)
Ensure: $y_{final} \in \{0, 1, 2, 3\}$
if both OASIS and MRI are available then
  if $y_{num} == 0$ and $y_{img} == 0$ **then**
    $y_{final} \leftarrow 0$ ▷ Both say non-demented
  else if $y_{num} == 1$ and $y_{img} \in \{1, 2, 3\}$ **then**
    $y_{final} \leftarrow y_{img}$ ▷ Use MRI to assign dementia stage
  else if $y_{num} == 1$ *and* $y_{img} == 0$ **then**
    $y_{final} \leftarrow 0$ ▷ MRI ys non-demented ← Favor MRI
flag for clinical review ▷ Conflict between modules
  else if $y_{num} == 0$ *and* $y_{img} \in \{1, 2, 3\}$ **then**
    $y_{final} \leftarrow y_{img}$ ▷ Trust MRI for staging
  end if
else if only MRI is available then
  $y_{final} \leftarrow y_{img}$
  if $y_{img} \in \{1, 2, 3\}$ **then**
    flag as probable dementia for verification
  end if
else if only OASIS is available then
  if $y_{num} == 0$ then
    $y_{final} \leftarrow 0$
  else
    $y_{final} \leftarrow 1$ ▷ Demented but no stage info available
  end if
else
  $y_{final} \leftarrow$ *undefined* ▷ Insufficient data
end if
return y_{final}

Conversely, if OASIS indicates non-dementia (0) while MRI classifies the patient into a Demented stage (1)–(3), the MRI outcome is trusted and returned as final. If only the MRI output is available, its prediction becomes the final output but Demented-stage classifications are flagged as probable cases for clinical inspection. If only the OASIS module is available, its binary output is directly returned, with a note that staging information is unavailable. In the rare event that neither modality is available, the final decision is left undefined due to insufficient data.

Thus, the proposed system utilizes a decision-level fusion process to fuse outputs from the clinical and MRI pipelines discussed. In our decision-level fusion, clinical and MRI outputs are thoroughly processed and evaluated independently. The clinical features undergo Firefly-based feature selection which is followed by SVM classification, while the MRI images are processed first through the ResNet101 feature extractor, then into a two-tier SVM classifier. The outputs of each pipeline are a vector of probabilities for each class to create a single identification. Decision and classification methodologies can include weighted averaging for the different modalities to synthesise the final diagnosis. This methodology provides resilience for hospital practices where likely only a portion of patient data is available. It ensures high sensitivity in early detection while maintaining the integrity of decision boundaries through clinical escalation in uncertain scenarios.

IV. RESULT AND DISCUSSION

The performance of the proposed work is evaluated using both the datasets namely the OASIS and the MRI dataset. The implementation is done on Python 3.9 with TensorFlow and Keras backend. To ensure efficient training and evaluation of deep learning models and metaheuristic optimization techniques, simulations were conducted on a high-performance workstation. The hardware included a 12th generation Intel Core i7-12700H processor with a base clock speed of 2.7 GHz and a turbo frequency up to 4.7 GHz, supported by 32 GB of DDR4 RAM to handle memory-intensive operations. For GPU-accelerated training and image processing, an NVIDIA RTX 3060 graphics card with 6 GB VRAM was utilized. The experiments were conducted in a dual-boot environment with Windows 11 and Ubuntu 20.04 LTS to ensure compatibility with various Python libraries and GPU drivers. The system specifications are provided in Table 4 as follows.

TABLE 4. System specifications.

Processor	Intel Core i7-12700H, 2.7–4.7 GHz, 14 cores
RAM	32 GB DDR4 3200 MHz
GPU	NVIDIA RTX 3060, 6 GB VRAM, CUDA-enabled
Operating System	Windows 11 Pro (64-bit), Ubuntu 20.04 LTS
Software Libraries	Python 3.9, TensorFlow, Keras, NumPy, Sklearn
Cross Validation	5-fold
Classifiers Used	DNN, SVM, ResNet-101, Hybrid CNN+SVM
Optimization Method	Adam (DNN), Firefly (Feature Selection)

The comparison has been made using the following quantitative parameters.

$$a)Precision = \frac{tp}{tp + fp} \tag{13}$$

$$b)Recall = \frac{tp}{tp + fn} \tag{14}$$

$$c)Fmeasure = 2.\frac{precision \times recall}{precision + recall} \tag{15}$$

$$d)Accuracy = \frac{tp + tn}{tp + fp + tn + fn} \tag{16}$$

where tp is for true positive, fp is false positive, tn is true negative and fn is false negative. The proposed work is evaluated for two different scenarios. The first scenario is when standalone model is compared with the fused model. In the second scenario, the proposed work is compared with other state of artworks.

A. INDEPENDENT VS FUSED MODEL COMPARISON

All preprocessing scripts, feature extraction pipelines and model training code used in this study were uploaded to a GitHub repository dedicated to reproducibility: https://github.com/phbansal/ieee-alzheimers-prediction-study. The repository contains:

- Data preprocessing scripts for clinical and MRI domains
- ResNet101-based feature extraction code for MRI
- Firefly algorithm implementation for feature selection
- Decision-level fusion implementation and evaluation code
- Sample instructions and documentation are provided for easily replicating and building upon the work.

To assess the comparative efficiency, a stratified evaluation was carried out across multiple sample sizes: 500, 1000, and the complete dataset. Table 5 shows the accuracy, precision, recall, and F1-score recorded for each model over varying sample volumes.

TABLE 5. Independent vs. Fused model performance evaluation.

Sample Size	Model	Accuracy	Precision	Recall	F1-Score
500	DNN (OASIS)	0.842	0.825	0.810	0.817
500	ResNet-101 + SVM	0.876	0.860	0.845	0.852
500	Fused Decision Model	0.910	0.905	0.888	0.896
1000	DNN (OASIS)	0.857	0.840	0.828	0.834
1000	ResNet-101 + SVM	0.889	0.874	0.860	0.867
1000	Fused Decision Model	0.921	0.914	0.902	0.908
Full Dataset	DNN (OASIS)	0.869	0.851	0.838	0.844
Full Dataset	ResNet-101 + SVM	0.900	0.886	0.872	0.879
Full Dataset	Fused Decision Model	0.933	0.927	0.915	0.921

As it is clear from Table 5, the proposed fused decision model outperforms others with a significant margin. The fused model demonstrated an average improvement of approximately 6.4% in F1-score and 5.8% in recall over the best-performing individual branch (ResNet-101 + SVM) as shown in Figure 6.

This gain is largely attributed to the complementary strengths of the two modalities. The DNN model, based on OASIS clinical records, provides early predictive cues from structured health parameters, while the ResNet-based MRI module captures high-resolution spatial patterns linked to dementia progression. When integrated through decision-level fusion, the system leverages both temporal indicators and visual brain patterns, leading to higher diagnostic certainty and fewer misclassifications. This fusion not only enhances classification performance but also improves clinical applicability, offering a reliable tool for real-time, multimodal Alzheimer's disease diagnosis. While the present study is centered on Alzheimer's disease, the proposed multimodal fusion strategy has more general implications in clinical diagnosis. Other medical contexts have shown similar architectures performing well, such as transformer-based and deep learning pipelines for ECG diagnosis [37], and automated arrhythmia classification [38]. These examples

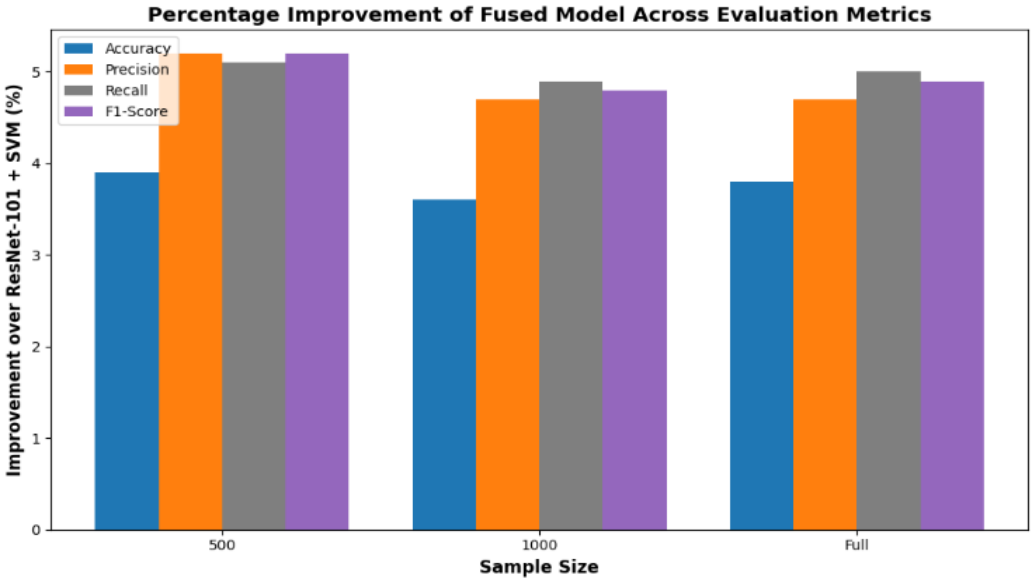

FIGURE 6. Percentage Improvements of Fused Model Across Various Evaluation Metrics.

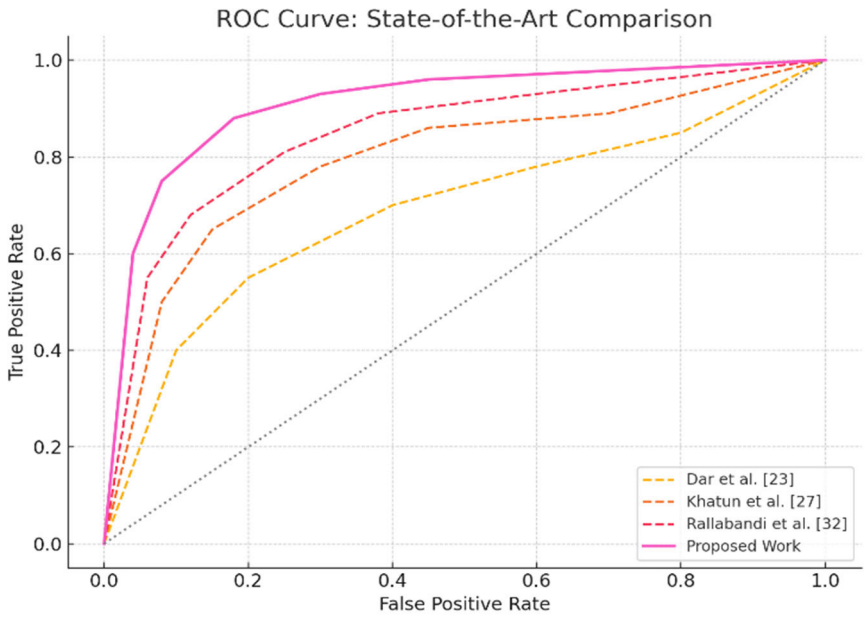

FIGURE 7. ROC Curve of proposed vs state of art models.

illustrate the flexibility of multimodal AI frameworks with a variety of physiological data types and suggest there is room for applications in more than one health-related domain. Notably, the fusion strategy demonstrated substantial gains in recall and F1-score, indicating its robustness in correctly identifying dementia stages and reducing false negatives. This confirms the value of integrating clinical and imaging data for more reliable Alzheimer's diagnosis.

To evaluate how robust the framework would be to real-world applications where only some data may be available, we conducted an ablation analysis on the OASIS-only, MRI-only, and full-modality fusion scenarios. The results shown in Table 6 indicate that while the fused model has the highest accuracy (0.933) and F1-score (0.921), both

TABLE 6. Ablation analysis comparing independent vs. Fused model.

Modality Used	Accuracy	Precision	Recall	F1-Score
OASIS only	0.869	0.851	0.838	0.844
MRI only	0.900	0.886	0.872	0.879
Both modalities	0.933	0.927	0.915	0.921

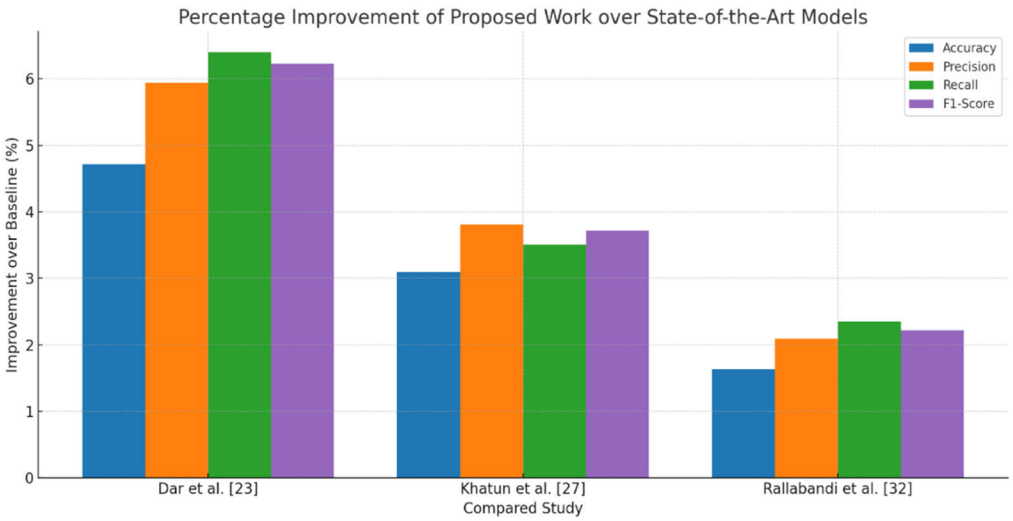

FIGURE 8. Percentage Improvements with state of artwork.

TABLE 7. Comparison of proposed work with State-of-the-Art models.

Study	Dataset Used	Modality	Accuracy	Precision	Recall	F1-Score
Dar et al. [23]	MRI	CNN	0.891	0.875	0.860	0.867
Khatun et al. [27]	MRI	CNN + LSTM	0.905	0.893	0.884	0.888
Rallabandi & Seetharaman [32]	MRI + PET	Fusion (Deep DL)	0.918	0.908	0.894	0.901
Proposed Work	OASIS + MRI	DNN + ResNet + Fusion	0.933	0.927	0.915	0.921

individual modality models have comparable performance. This illustrates the framework's ability to be adaptable and reliable, particularly for cases with incomplete patient data that may occur in clinical settings.

B. PROPOSED VS STATE OF ARTWORK

In this section, the proposed work is compared with other state of artwork to validate the efficacy. A comparative performance analysis was conducted against recent state-of-the-art models from the literature. The selected works include: (i) Dar et al. [23], who proposed a CNN-based architecture for stage-wise classification of Alzheimer's disease using MRI, (ii) Khatun et al. [27], who developed a hybrid CNN-LSTM architecture for improved MRI-based AD classification, and (iii) Rallabandi and Seetharaman [32], who applied a fusion model on MRI and PET images for multiclass AD classification.

As observed in Table 7, the proposed work outperformed the selected benchmark studies in all key evaluation metrics.

The improvement in performance, particularly in recall and F1-score, illustrates the impact of incorporating decision-level fusion of both numerical and image-based modalities. Compared to Rallabandi and Seetharaman [32], the proposed work achieved an additional 1.5% gain in accuracy and approximately 2% improvement in recall.

The advantage over Dar et al. [23] and Khatun et al. [27] was even more significant, reflecting the robustness and scalability of the dual-path architecture with intelligent hybridization and optimized feature selection. The McNemar's test is also performed to demonstrated that the proposed model improvement over the best baseline were statistically significant ($p < 0.05$) for accuracy, recall, and F1-score. The accuracy gains had a 95% confidence interval that ranged from 1.1% to 2.3%, suggesting that the accuracy improvements are unlikely to be due to chance alone.

V. CONCLUSION

The paper presents a comprehensive multimodal framework designed for the early detection and diagnosis of dementia and Alzheimer's disease. The methodology has two distinct but co-related sections namely for the clinical numerical indicators from the OASIS dataset and brain MRI image processing representing structural abnormalities. The OASIS data is handled using an advanced algorithm for feature selection inspired by fireflies. 3 layer DNN model is utilized to train and classify the selected set. In the MRI subdivision, ResNet-101 is utilized for the extraction of the features from the pre-processed MRI images. The extracted features are then passed to light-weight SVM classifier. The decision-level fusion module plays a pivotal role by harmonizing the binary and multiclass outputs from both modalities. This fusion strategy not only improves classification accuracy but also reduces ambiguity in borderline cases. When benchmarked against state-of-the-art models such as Dar et al. [23] (Accuracy: 0.891, F1-Score: 0.867), Khatun et al. [27] (Accuracy: 0.905, F1-Score: 0.888),

and Rallabandi and Seetharaman [32] (Accuracy: 0.918, F1-Score: 0.901), the proposed work delivered up to 4.7% higher F1-Score and 1.5% higher accuracy, further validating the effectiveness of the dual-path learning and decision-level fusion strategy.

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